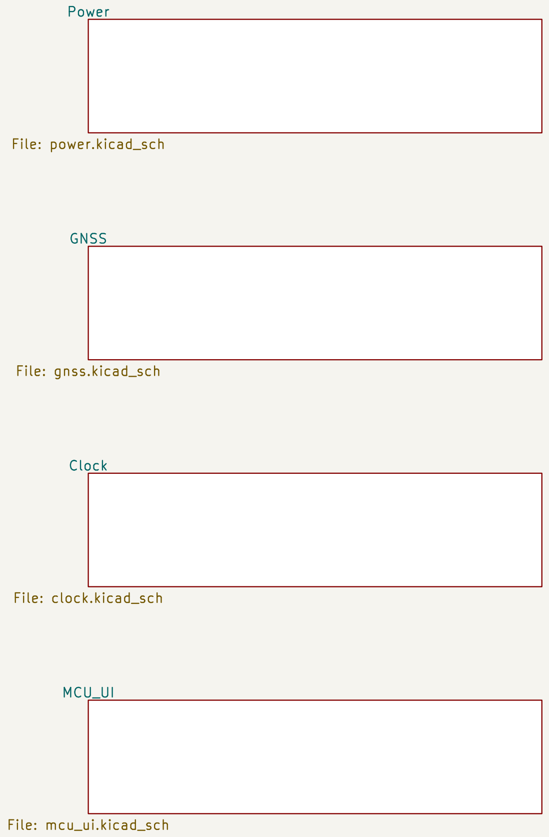

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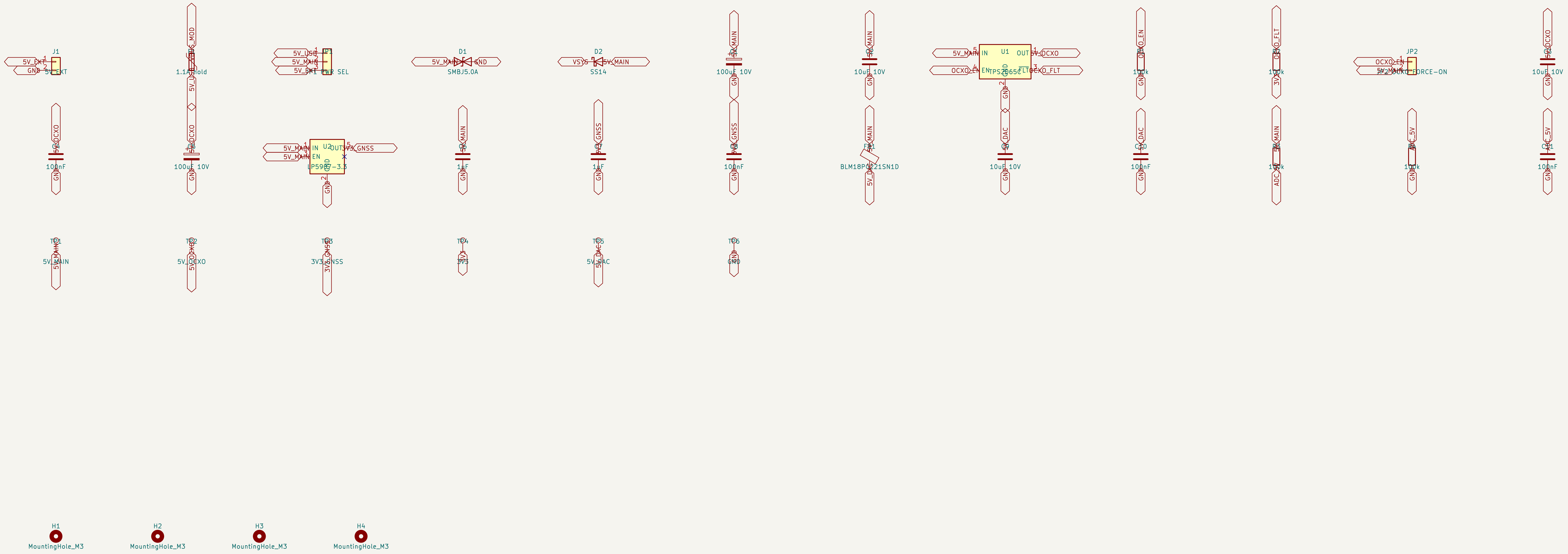
Title:

Size: A3 Date: **Rev:**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 Id: 1/5

POWER: 5V_USB (VBUS) / 5V_EXT -> JP1 -> 5V_MAIN -> TVS, OCXO switch, 3V3_GNSS LDO, ferrite-filtered 5V_DAC

JP1: 1-2 = USB power, 2-3 = external 5V. Module VSYS fed from 5V_MAIN through Schottky (D2). VBUS pad is direct USB VBUS.



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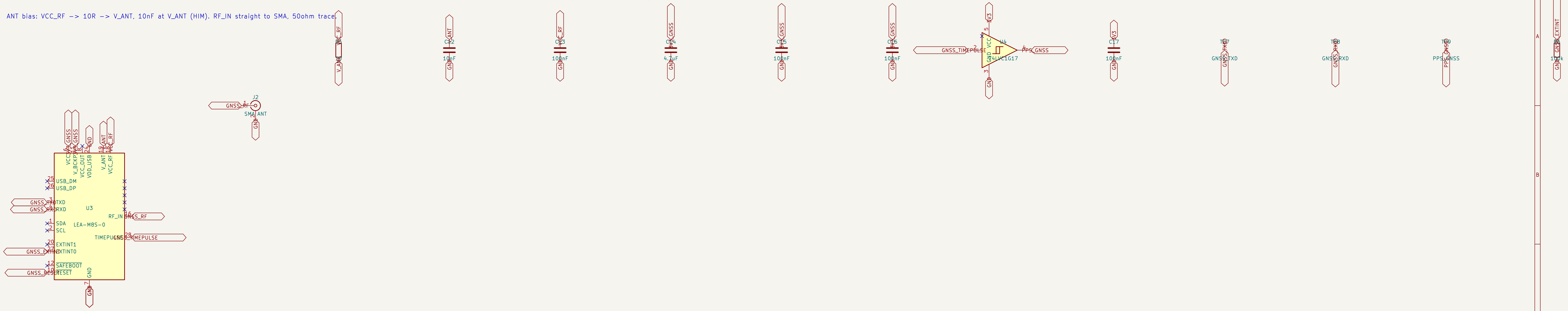
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 2/5

GNSS: u−blox LEA−M8S on 3V3_GNSS. UART to MCU, TIMEPULSE −> 74LVC1G17 −> GP6. USB not used (VDD_USB=GND).

ANT bias: VCC_RF −> 10R −> V_ANT, 10nF at V_ANT (HIM). RF_IN straight to SMA, 50ohm trace.



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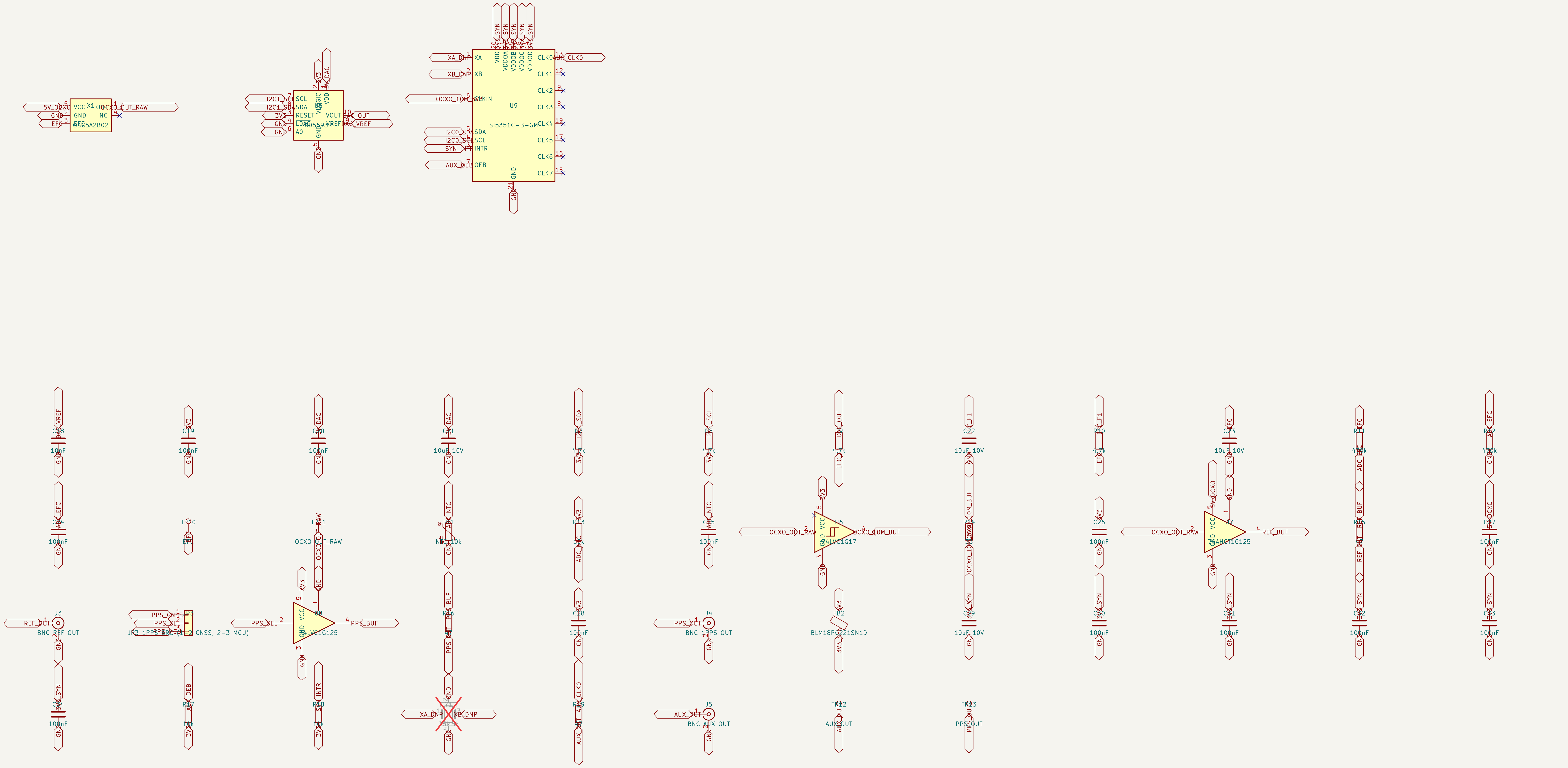
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4
Id: 3/5

CLOCK: OCXO + EFC DAC, 10MHz buffers, 1PPS OUT mux, Si5351C AUX synth (CLKIN = OCXO). I2C1: DAC 0x4C. I2C0: Si5351 0x60 + OLED 0x3C.

OCXO pin geometry unverified: placeholder 1x5 header, pad N = OCXO pin N (datasheet bottom view). Verify before layout.



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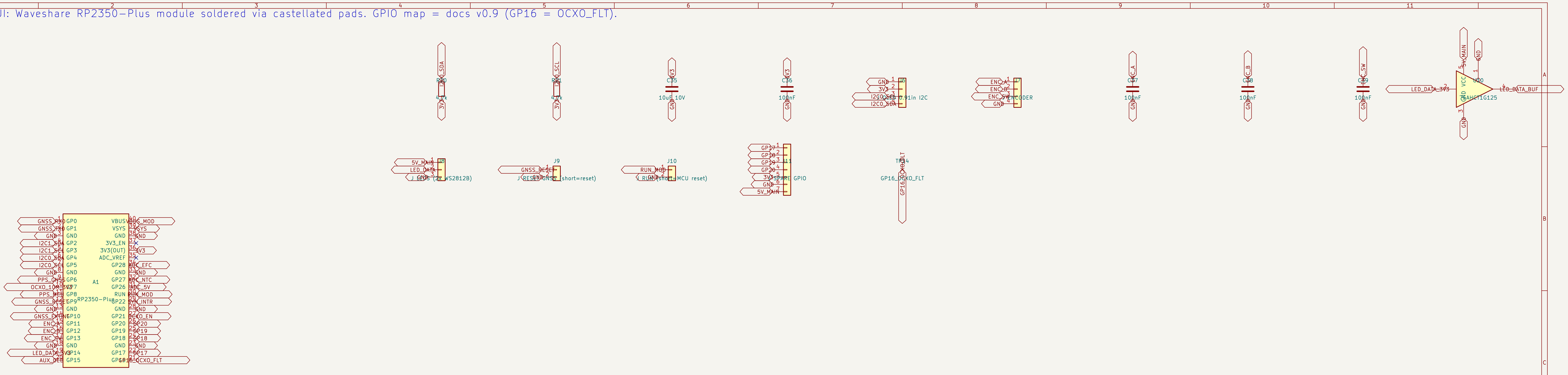
Title:

Size: A2
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 4/5

MCU + UI: Waveshare RP2350-Plus module soldered via castellated pads. GPIO map = docs v0.9 (GP16 = OCXO_FLT).



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KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 5/5